

We consider a set of order pair of integers

$$Q = \{(x, y) \mid x, y \in \mathbb{Z} \text{ and } y \neq 0\}.$$

The first element x of a pair $(x, y) \in Q$ is called the **numerator**, and the second element y is called the **denominator**. Elements of Q are called **numerator-denominator pairs**.

Next, we introduce an equivalent relation in Q :

Definition 3. We say that two elements (x, y) and (z, w) of Q are **rationally equivalent** if $x \cdot w = y \cdot z$, i.e.,

$$(x, y) \sim (z, w) \iff x \cdot w = y \cdot z$$

where $\cdot$ is the multiplication defined on $\mathbb{Z}$.

Proposition 5. The relation $\sim$ is an equivalence relation on the set Q .

Proof. Exercise. The reflexive and symmetry laws involve using the commutative law for $\mathbb{Z}$. The transitivity law requires multiplying an equation by a constant, performing basic manipulations, and using the cancellation law (for a non-zero element of $\mathbb{Z}$). □

Definition 4. If $(x, y) \in Q$ then $[(x, y)]$ denotes the equivalence class containing (x, y) under the above equivalence relation. We define

$$\mathbb{Q} = Q / \sim.$$

In other words,

$$\mathbb{Q} = \left\{ [(x, y)] \mid x, y \in \mathbb{Z} \text{ with non-zero } y \right\} / \sim.$$

2.1 Addition and Multiplication.

We define the usual operations on $\mathbb{Q}$ as follows: let $[(x, y)]$ and $[(z, w)]$ be elements of Q . Then addition +

$$[(x, y)] + [(z, w)] = [(xw + yz, yw)] \tag{0.7}$$

multiplication $\cdot$

$$[(x, y)] \cdot [(z, w)] = [(xz, yw)] \tag{0.8}$$

Similar to the case for the integers, these definitions involve equivalence classes, so we must show that they are well-defined. We leave those justifications as an exercise. For example, to verify that *addition on $\mathbb{Q}$ is well-defined*:

Proof. We want to show that if $[(x, y)] = [(x', y')]$ and $[(z, w)] = [(z', w')]$ then

$$[(xw + yz, yw)] = [(x'w' + y'z', y'w')].$$

In other words, suppose $(x, y) \sim (x', y')$ and $(z, w) \sim (z', w')$. We must show $(xw + yz, yw) \sim (x'w' + y'z', y'w')$. By definition of $\sim$, we need to show that

$$(xw + yz)(y'w') = (x'w' + y'z')(yw).$$

Similarly, we can state the following propositions without proof:

Proposition 6.

- i). *Multiplication on $\mathbb{Q}$ is well-defined.*
- ii). *Addition in $\mathbb{Q}$ is commutative.*
- iii). *Multiplication in $\mathbb{Q}$ is commutative.*

Proposition 7. *For any $[(x, y)] \in \mathbb{Q}$,*

$$[(x, y)] + [(0, 1)] = [(x, y)] \quad \text{and} \quad [(x, y)] \cdot [(1, 1)] = [(x, y)].$$

This proposition implies

- $[(0, 1)] \in \mathbb{Q}$ is the identity for addition;
- $[(1, 1)] \in \mathbb{Q}$ is the identity for multiplication.

2.2 Division.

Using definition (0.8), we can easily compute that

$$[(x, y)] \cdot [(y, x)] = [(1, 1)] \quad \text{for any non-zero integers } x, y.$$

This implies that

for any non-zero $x, y \in \mathbb{Z}$, the multiplication inverse of $[(x, y)]$ is $[(y, x)]$,

and we denote the multiplication inverse $[(y, x)]$ as $[(x, y)]^{-1}$.

The existence of the multiplication inverse leads us to introduce the **division** operation: for nonzero $z, w \in \mathbb{Z}$, we define

$$[(x, y)] / [(z, w)] := [(x, y)] \cdot [(z, w)]^{-1}. \tag{0.9}$$

2.3 Identifying $\mathbb{Z}$ as a subset of $\mathbb{Q}$.

The identification can be done through the canonical embedding $\mathbb{Z} \rightarrow \mathbb{Q}$ defined by the rule

$$a \mapsto [(a, 1)].$$

If we identify $a \in \mathbb{Z}$ with its image in $\mathbb{Q}$, then we can think of $\mathbb{Z}$ as a subset of $\mathbb{Q}$. So from now on, if $a \in \mathbb{Z}$, we will think of a and $[(a, 1)]$ as being the same element of $\mathbb{Q}$. By this convention, $0 \in \mathbb{Z}$ is identified with $[(0, 1)]$, and $1 \in \mathbb{Z}$ is identified with its image $[(1, 1)]$. By a propositions above,

$0 \approx [(0, 1)]$ is the additive identity and $1 \approx [(1, 1)]$ is the multiplicative identity as expected.

Since we now think of $\mathbb{Z}$ as a subset of $\mathbb{Q}$, we have to be careful with $+$ and $\cdot$ in $\mathbb{Z}$. We defined these operations for $\mathbb{Z}$ in one way previously, and then defined them for $\mathbb{Q}$ in a different way. Do we get the same answer for integers $a, b \in \mathbb{Z}$ as for the corresponding elements $[(a, 1)]$ and $[(b, 1)]$ in $\mathbb{Q}$? The answer is yes since

$$a + b \approx [(a, 1)] + [(b, 1)] = [(a + b, 1)] \approx a + b$$

and

$$a \cdot b \approx [(a, 1)] \cdot [(b, 1)] = [(a \cdot b, 1 \cdot 1)] = [(ab, 1)] \approx a \cdot b.$$

Likewise for the additive inverse:

$$-[(a, 1)] = [(-a, 1)].$$

This equality shows that if a is identified with $[(a, 1)]$, then the definitions of additive inverse, either as an integer or as a fraction, gives the same result. We summarize the above observations as follows.

Proposition 8. *Consider $\mathbb{Z}$ as a subset of $\mathbb{Q}$. Then the addition, multiplication, and additive inverse operators on $\mathbb{Q}$ extend the corresponding operators on $\mathbb{Z}$.*

2.4 Fraction notation of $\mathbb{Q}$:

Using the identification, we can also rewrite the multiplication inverse: for any nonzero $a \in \mathbb{Z}$, $a = [(a, 1)]$, its multiplication inverse is

$$[(1, a)] = [(a, 1)]^{-1} := a^{-1} \text{ or } \frac{1}{a}.$$

Next, for any $a, b \in \mathbb{Z}$ with $b \neq 0$, it follows from the multiplication that

$$[(a, b)] = [(a, 1)] \cdot [(1, b)].$$

Using the identification, this simply gives

$$[(a, b)] = \frac{a}{b}.$$

So, we can also write the rational number $[(a, b)]$ using the fractional notation $\frac{a}{b}$. Therefore,

$$\mathbb{Q} = \left\{ \frac{m}{n} : m, n \in \mathbb{Z}; n \neq 0 \right\}$$

That is, if r is an element in the set $\mathbb{Q}$, then r can be written as the quotient of two integers. We require that m and n have no common factor, other than ± 1 . Then,

for any $r \in \mathbb{Q}$, there are unique $m, n \in \mathbb{Z}$ with $(m, n) = \pm 1$ such that $r = \frac{m}{n}$.

Using this notation of fraction, the two operations (0.7) and (0.8) defined on $\mathbb{Q}$ can now be rewritten as

+ (addition):

$$\frac{m_1}{n_1} + \frac{m_2}{n_2} := \frac{m_1 n_2 + m_2 n_1}{n_1 n_2};$$

• (multiplication):

$$\frac{m_1}{n_1} \cdot \frac{m_2}{n_2} := \frac{m_1 m_2}{n_1 n_2}.$$

Moreover, we also have

- $0 \in \mathbb{Q}$ is the identity for addition, that is

$$q + 0 = 0 + q = q \text{ for any } q \in \mathbb{Q}.$$

- $1 \in \mathbb{Q}$ is the identity for multiplication, that is

$$q \cdot 1 = 1 \cdot q = q \text{ for any } q \in \mathbb{Q}.$$

2.5 $\mathbb{Q}$ is an ordered field.

We first recall some definitions from algebra.

Definition 5. Let F be a set. We say that F is a **field** if there are two operations called *addition* “+” and *multiplication* “ $\cdot$ ” defined on F such that the following axioms hold

- (A) Axioms for addition:
- A1). If $a, b \in F$, then $a + b \in F$; (*closed under “+”*)
 - A2). $a + b = b + a$ for all $a, b \in F$; (*commutative law*)
 - A3). $a + (b + c) = (a + b) + c$ for all $a, b, c \in F$; (*associative law*)
 - A4). There is $0 \in F$ such that $a + 0 = a$ for all $a \in F$; ($\exists$ *identity for “+”*)
 - A5). For each $a \in F$, there is an element $-a \in F$ such that $a + (-a) = 0$; ($\exists$ *inverse for “+”*)
- (M) Axioms for multiplication:
- M1). If $a, b \in F$, then $a \cdot b \in F$; (*closed under “ $\cdot$ ”*)
 - M2). $a \cdot b = b \cdot a$ for all $a, b \in F$; (*commutative law*)
 - M3). $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in F$; (*associative law*)
 - M4). There is $1 \in F$ such that $a \cdot 1 = a$ for all $a \in F$; ($\exists$ *identity for “ $\cdot$ ”*)
 - M5). For each $a \neq 0$, there is an element $a^{-1} \in F$ such that $a \cdot a^{-1} = 1$; ($\exists$ *inverse for “ $\cdot$ ”*)
- DL). $a \cdot (b + c) = a \cdot b + a \cdot c$ for all $a, b, c \in F$.

Using the definitions (0.7) and (0.8) together with the propositions, we can verify that

Proposition 9. $(\mathbb{Q}, +, \cdot)$ is a field.

2.6 $\mathbb{Q}$ is an ordered field.

Indeed, the rational number set $\mathbb{Q}$ also has an *order structure*.

Definition 6. Let F denote a set. F is called an **ordered field** if F is a field and also has an order “ $<$ ” such that this order structure is preserved by the algebraic operations “+” and “ $\cdot$ ”, that is,

- O3). If $a < b$, then $a + c < b + c$;
- O4). If $a < b$ and $0 < c$, then $a \cdot c < b \cdot c$.

For $\mathbb{Q}$, we can define an order “ $<$ ” as following: for any $q_1 = \frac{m_1}{n_1}$ and $q_2 = \frac{m_2}{n_2}$ with $m_1, m_2 \in \mathbb{Z}$ and $n_1, n_2 \in \mathbb{Z}_+$, we define

$$q_1 < q_2 \iff m_1 n_2 < m_2 n_1,$$

where ‘ $<$ ’ on the right hand side is the order on $\mathbb{Z}$ we introduced earlier.

With this definition, we can verify that:

Proposition 10. $(\mathbb{Q}, +, \cdot, <)$ is an ordered field.

2.7 The Incompleteness of $\mathbb{Q}$.

A natural question arises:

can we solve general algebraic equations like $x^2 = 2$ over $\mathbb{Q}$?

Theorem 1. *There is no $x \in \mathbb{Q}$ that satisfies $x^2 = 2$.*

Proof. We will argue by contradiction. Suppose there is such an $x \in \mathbb{Q}$. Then, there exist $m, n \in \mathbb{Z}$ such that

$$x = \frac{m}{n}, \quad (m, n) = 1$$

This implies $m = x \cdot n$, and squaring both sides gives $m^2 = x^2 \cdot n^2 = 2n^2$. Therefore, $2 \mid m^2$, which implies that $2 \mid m$. We may write $m = 2k$ for some integer k . Substituting this back, we have

$$4k^2 = 2n^2 \implies 2k^2 = n^2$$

This implies $2 \mid n^2$, and hence $2 \mid n$. Therefore, m and n share a common factor of 2, which contradicts our initial assumption that $(m, n) = 1$. □

Remark 4. *From our experience with the constructions of $\mathbb{Z}$ from $\mathbb{N}$, and $\mathbb{Q}$ from $\mathbb{Z}$, the rough idea is to define $\mathbb{R}$ to be the set with its elements generated from $\mathbb{Q}$. In the next lecture, we will give the brief idea of two constructions: Dedekind cuts (Dedekind) and Cauchy sequences (Cantor).*

3. Some properties of ordered field (reading material; skipped in the lecture).

Theorem 2. *If $(F, +, \cdot)$ is a field, then, for any $a, b, c \in F$, we have*

- (i) $a + c = b + c$ implies $a = b$;
- (ii) $a \cdot 0 = 0$ for all $a \in F$;
- (iii) $(-a) \cdot b = -a \cdot b$ for all $a, b \in F$;
- (iv) $(-a) \cdot (-b) = a \cdot b$ for all $a, b \in F$;
- (v) $a \cdot c = b \cdot c$ and $c \neq 0$ implies $a = b$;
- (vi) $a \cdot b = 0$ implies either $a = 0$ or $b = 0$.

Proof: (Keep in mind that the operations we can use in the proof are the properties A1)-A4), M1)-M4) and DL).)

- (i) $a + c = b + c$ implies $(a + c) + (-c) = (b + c) + (-c)$, so by A1), we have

$$a + [c + (-c)] = b + [c + (-c)].$$

By A4), this reduces to $a + 0 = b + 0$ and therefore $a = b$ by A3).

- (ii) We use A3) and DL) to obtain

$$a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0,$$

so, $0 + a \cdot 0 = a \cdot 0 + a \cdot 0$ by A3) again. By (i), we conclude $0 = a \cdot 0$.

(iii) Since $a + (-a) = 0$, we have

$$a \cdot b + (-a) \cdot b = [a + (-a)] \cdot b = 0 \cdot b = 0 = a \cdot b + (-a \cdot b)$$

From (i), we obtain $(-a) \cdot b = -(a \cdot b)$.

(iv) Using (iii) and M2), we have

$$(-a) \cdot (-b) = -(a \cdot (-b)) = -((-b) \cdot a) = -(-(b \cdot a)) = a \cdot b.$$

(v) $a \cdot c = b \cdot c$ and $c \neq 0$ implies $(a \cdot c) \cdot c^{-1} = (b \cdot c) \cdot c^{-1}$. By M1), we have

$$a \cdot (c \cdot c^{-1}) = b \cdot (c \cdot c^{-1}).$$

By M4), this reduces to $a \cdot 1 = b \cdot 1$ and therefore $a = b$ by M3).

(vi) If $a \neq 0$ and $a \cdot b = 0$, we have $a^{-1} \cdot (a \cdot b) = a^{-1} \cdot 0 = 0$. Then, $b = (a^{-1} \cdot a) \cdot b = 0$.

□

Theorem 3. If $(F; +, \cdot, \leq)$ is an ordered field, then we have, for any $a, b, c \in F$,

- (i) If $a \leq b$, then $-b \leq -a$;
- (ii) If $a \leq b$ and $c \leq 0$, then $bc \leq ac$;
- (iii) If $0 \leq a$ and $0 \leq b$, then $0 \leq ab$;
- (iv) $0 \leq a^2$ for all $a \in F$;
- (v) $0 < 1$;
- (vi) If $0 < a$, then $0 < a^{-1}$;
- (vii) If $0 < a < b$, then $0 < b^{-1} < a^{-1}$.

Remark 5.

- $a < b$ means $a \leq b$ and $a \neq b$.
- We denote $a \geq b$ if $b \leq a$.

Proof of the theorem:

(iv) If $0 \leq a$, then $0 \leq a \cdot a$ by (iii) and hence $0 \leq a^2$. If $a \leq 0$, then $0 \leq -a$ by (i). Now using what we just proved case, we have $0 \leq (-a)^2 = a^2$.

(vii) If $0 < a < b$, then $0 < a \cdot a^{-1} < b \cdot a^{-1}$ which implies $1 < b \cdot a^{-1}$. Therefore,

$$0 < b^{-1} < b^{-1} \cdot (b \cdot a^{-1}) = (b^{-1} \cdot b) \cdot a^{-1} = a^{-1}.$$

(i) - (iii), (v) and (vi) are left as exercise.

LECTURE 3: REAL NUMBER $\mathbb{R}$

In the previous section, we discussed the rational number set $\mathbb{Q}$ and its important properties as an ordered field. Moreover, we also have

Proposition 11. *If $r \in \mathbb{Q}$, then r is a solution of a linear equation with integer coefficients.*

Proof: If $r \in \mathbb{Q}$, then $r = \frac{m}{n}$ for some integers m and n . Multiplying n on both sides, we have $n \cdot r = m$. Subtracting m on both sides, we obtain $n \cdot r - m = 0$. $\square$

This proposition indicates that we can solve linear equations with integer (or rational) coefficients over $\mathbb{Q}$. A natural question is: how about more general algebraic equations like quadratic equation? The example $x^2 = 2$ shows that the answer is no. Therefore, we should further enlarge our number system.

1. Construction of $\mathbb{R}$.

To enlarge the rational number set $\mathbb{Q}$, we would like to get a larger set satisfying

- including $\mathbb{Q}$ as a subset;
- keeping the ordered field structure.

From our experience with the constructions of $\mathbb{Z}$ from $\mathbb{N}$, and $\mathbb{Q}$ from $\mathbb{Z}$, the rough idea is to define $\mathbb{R}$ to be the set with its elements generated from $\mathbb{Q}$. Here, we give the brief idea of two constructions: **Dedekind cuts** (Dedekind) and **Cauchy sequences** (Cantor), both published in 1872.

1.1 Dedekind cuts.

Rough idea: the elements of $\mathbb{R}$ will be some *special* subsets of $\mathbb{Q}$, called *cuts*. A cut is, by definition, any set $\alpha \subseteq \mathbb{Q}$ with the following three properties:

- (I) α is not empty, and $\alpha \neq \mathbb{Q}$; (i.e. α is a proper subset)
- (II) If $p \in \alpha, q \in \mathbb{Q}$, and $q < p$, then $q \in \alpha$; (i.e. α is closed downwards)
- (III) If $p \in \alpha$, then $p < r$ for some $r \in \alpha$. (i.e. α contains no greatest element)

We define

$$\mathbb{R} := \left\{ \alpha \mid \alpha \subset \mathbb{Q} \text{ is a cut} \right\}$$

Next, we can define the order and algebraic operations on $\mathbb{R}$ and prove that it is an ordered field. For example, we can define

- $\mathbb{Q} \subset \mathbb{R}$: we can embed $\mathbb{Q}$ into $\mathbb{R}$ by identifying the rational number r with the cut

$$r^* := \{x \in \mathbb{Q} : x < r\}.$$

Indeed, using the following definition of the **order** on $\mathbb{R}$, we can easily check that the above map $r \rightarrow r^*$ is injective.

- **Oder:** for any $\alpha, \beta \in \mathbb{R}$,

$$\alpha < \beta \iff \alpha \subset \beta;$$

- **Addition:** for any $\alpha, \beta \in \mathbb{R}$,

$$\alpha + \beta := \{p \in \mathbb{Q} : \exists a \in \alpha \text{ and } \exists b \in \beta \text{ such that } p < a + b\};$$

The strict inequality in the definition is useful because it guarantees that the result is again a Dedekind cut. Equivalently, since α, β are downward closed, we can also define

$$\alpha + \beta := \{a + b : a \in \alpha, b \in \beta\}.$$

One can verify that this addition operation satisfies the properties A1) - A5) in the definition of *field*. In particular, *the identity for +* is given by the cut

$$0^* := \{p \in \mathbb{Q} : p < 0\};$$

and the additive inverse of a cut α is defined as (in textbook)

$$-\alpha := \{p \in \mathbb{Q} : \exists r \in \mathbb{Q}_+ \text{ such that } -p - r \notin \alpha\}.$$

Equivalently, we can also define $-\alpha$ as

$$-\alpha := \{p \in \mathbb{Q} : p < -a \text{ for some } a \notin \alpha\}.$$

- Intuitive idea is that the inverse should be the negative of the upper set $\mathbb{Q} \setminus \alpha$;
- the complement contains the boundary point and one needs to remove it to get a cut.
- for example: for $\alpha = \{q \in \mathbb{Q} : q < 3\}$, $\mathbb{Q} \setminus \alpha = \{p \in \mathbb{Q} : p \geq 3\}$ and

$$\begin{aligned} -\alpha &= \{p \in \mathbb{Q} : p < -a \text{ for some } a \in \mathbb{Q} \setminus \alpha\} \\ &= \{p \in \mathbb{Q} : p < -a \text{ for some } a \geq 3\} \\ &= \{p \in \mathbb{Q} : p < -3\}. \end{aligned}$$

- **Multiplication:** for $\alpha, \beta \in \mathbb{R}$ and $0^* < \alpha, 0^* < \beta$, we define

$$\alpha \cdot \beta = \{p \in \mathbb{Q} : p < r \cdot s \text{ for some } 0 < r \in \alpha, 0 < s \in \beta\}.$$

$\alpha \cdot \beta$ for general $\alpha, \beta \in \mathbb{R}$ can be defined based on this (see Step 7 on page 20 in the textbook). For example, if $\alpha < 0^*$ and $\beta > 0^*$, we define $\alpha \cdot \beta := -[(-\alpha) \cdot \beta]$.

One can verify that this operation satisfies the properties M1) - M5) and DL) in the definition of *field*. In particular, *the identity for ·* is given by the cut:

$$1^* := \{p \in \mathbb{Q} : p < 1\}.$$

Finally, after long and tedious work, we can verify that

the set $\mathbb{R}$ with the above defined structures is an ordered field.

As an example of a Dedekind cut representing an irrational number, we may take the positive square root of 2. This can be defined by the set

$$\alpha = \{x \in \mathbb{Q} : x < 0 \text{ or } x^2 < 2\}.$$

We need to verify that $\alpha \in \mathbb{R}$ and

$$\alpha \cdot \alpha = 2^* := \{r \in \mathbb{Q} : r < 2\}.$$

Proof. To show $\alpha \in \mathbb{R}$, we need to check that α is a cut. (I) and (II) are easy to show. For (III), we need to prove that α has no greatest element in $\mathbb{Q}$, i.e. that for any positive rational x with $x^2 < 2$, there is a rational $y \in \mathbb{Q}$ such that

$$x < y \quad \text{and} \quad y^2 < 2.$$

Indeed, one can check that $y = x + \frac{2-x^2}{2+x}$ works since direct computation implies $y^2 - 2 = \frac{2(x^2-2)}{(x+2)^2} < 0$. Hence, $y > x$ and $y \in \alpha$.

Next, we need to check

$$\alpha \cdot \alpha = 2^*.$$

This is an equation of two sets. For this, we should prove both $\alpha \cdot \alpha \subseteq 2^*$ and $2^* \subseteq \alpha \cdot \alpha$.

- For $\alpha \cdot \alpha \subseteq 2^*$: taking any $p \in \alpha \cdot \alpha$, then, by the definition of multiplication of cuts, there exist $0 < a, b \in \alpha$ such that

$$p < a \cdot b.$$

Note that, $0 < a, b \in \alpha$ implies $a^2 < 2$ and $b^2 < 2$. Therefore, by the ordered field structure,

$$p^2 < a^2 \cdot b^2 < 4.$$

It follows that $p < 2$, and therefore, $p \in 2^*$.

- For $2^* \subseteq \alpha \cdot \alpha$: taking any $p \in 2^*$, that is, $p \in \mathbb{Q}$ and $p < 2$, we want to show that $p \in \alpha \cdot \alpha$. For that, we need to find $0 < a \in \alpha$, such that

$$p < a^2 < 2,$$

which is a direct consequence of the following more general lemma.

Lemma. For any two rational numbers $0 < a < c$, there exists $b \in \mathbb{Q}$ such that

$$a < b^2 < c.$$

Proof of Lemma. Suppose that there is no such b . Then, for each $n \in \mathbb{N}_+$, there is $m := m_n \in \mathbb{N}$ such that

$$\frac{m^2}{n^2} \leq a < c \leq \frac{(m+1)^2}{n^2} = \frac{m^2}{n^2} + \frac{2m+1}{n^2}.$$

This implies that, for any $n \in \mathbb{N}_+$, we have

$$c - a \leq \frac{2m+1}{n^2}$$

and therefore

$$m \geq \frac{(c-a)n^2 - 1}{2}$$

which gives

$$\frac{m^2}{n^2} \geq \frac{((c-a)n^2 - 1)^2}{4n^2} = \frac{(c-a)^2 n^2}{4} - \frac{c-a}{2} + \frac{1}{4n^2} \geq \frac{(c-a)^2 n^2}{4} - \frac{c}{2}$$

However, recall that $\frac{m^2}{n^2} \leq a$, so we have

$$a \geq \frac{(c-a)^2 n^2}{4} - \frac{c-a}{2}$$

and it implies

$$n^2 \leq \frac{2(a+c)}{(c-a)^2} \quad \text{for all } n \in \mathbb{N}_+.$$

This is impossible.

□

Remark 6. Students may still be curious: can we use Dedekind cut to construct every irrational number, like π, e ? Indeed, we do not really care about it. The essential purpose of Dedekind cuts is to demonstrate that there is a set of objects, defined only using the established properties of $\mathbb{Q}$, that satisfy the *complete ordered field axioms* - they “take the completion of the rationals” by cleverly characterizing the “holes”.